Muqi (Barry) Xu

+1 (217) 637-3442 | muqixu@outlook.com | linkedin.com/in/muqi-xu | muqixu.github.io

Education

University of Illinois Urbana-Champaign Expected Dec. 2026
Bachelor of Science in Computer Engineering GPA: 3.73/4.0

- **Relevant Coursework:** Computer Organization and Design, Applied Parallel Programming, Computer Systems Engineering, Digital Systems Laboratory, VLSI System Design, IoT and Cognitive Computing.

Carnegie Mellon University Expected May 2028
Master of Science in Electrical and Computer Engineering Starting Jan. 2027

Technical Skills

HDL & Programming: SystemVerilog, C/C++, Python, RISC-V Assembly, CUDA, Shell, Makefile
Digital Design: RTL Design, Pipelining, FSMs, Computer Architecture, Caches, Out-of-Order Execution, FPGA Prototyping
ASIC/FPGA Tools: Synopsys VCS/Verdi, Xilinx Vivado/Vitis, Cadence Virtuoso/Innovus, OpenRAM, Git
Verification & Automation: UVM, SVA, Functional Coverage, Waveform/Log Debug, Python Automation, Claude Code Skills

Selected Projects

Two-Way Superscalar Out-of-Order RISC-V Processor SystemVerilog, VCS/Verdi

- Designed a two-way superscalar **out-of-order RV32IM processor** with explicit register renaming (RAT/RRAT, free list, PRF), reservation stations, ROB, and split load/store queues.
- Widened fetch, rename, and commit to **two instructions per cycle** with intra-bundle dependency forwarding; built CDB arbitration for five functional units and integrated DesignWare mul/div.
- Built a **GShare predictor** (128-entry PHT, 7-bit GHR) and a dual-stack **return-address stack** with misprediction recovery, raising CoreMark IPC by **20%** and branch accuracy to 86–99%.
- Implemented **performance counters** to locate bottlenecks and drive architecture decisions: resized reservation stations toward ALU-heavy workloads under the area limit and exposed I-cache stalls.
- Designed a **2-stage pipelined I-cache** (1-cycle hit, 1 access/cycle vs. 0.5) from counter data, improving IPC up to **15%**; resolved critical paths to **575 MHz** and applied clock gating.
- Verified with **directed assembly tests** and a **random instruction generator** for **100% ISA functional coverage**, **SVA** on retirement, flush, and ROB invariants, and commit-log/Verdi debug.

Four-Way Set-Associative Cache and Memory Subsystem SystemVerilog, UVM, OpenRAM, VCS/Verdi

- Designed a **4-way, 16-set write-back/write-allocate cache** with 32-byte lines, byte-masked accesses, and 256-bit memory interface.
- Implemented FSM control for read/write hits, clean allocations, dirty evictions, and memory writebacks with per-line valid and dirty state.
- Designed a **Tree-PLRU replacement policy** and integrated OpenRAM-generated tag/data SRAM macros into the cache datapath.
- Verified with a reusable **UVM environment**: active CPU-side agent, reactive memory-side agent, and a self-checking reference model and scoreboard tracking data, tags, valid/dirty, and PLRU state.
- Wrote **20 directed/constrained-random sequences** (hits, clean/dirty misses, set conflicts, evictions) for **100% functional coverage**.

Five-Stage Pipelined RV32I Processor SystemVerilog, VCS/Verdi, Spike

- Designed a **5-stage in-order RV32I pipeline** with struct-based stage registers, valid-bit flushing, and global memory stalls.
- Implemented **full data forwarding** (MEM/WB to EX, WB to MEM, transparent register file), load-use hazard detection, and static not-taken prediction with misprediction flush.
- Folded the pipelined memory's 1-cycle latency into the EX/MEM boundary; generated byte-masked, aligned loads/stores.
- Met **500 MHz** timing within a **20,000** $\mu\mathbf{m}^2$ area budget, sustaining **0.99 IPC** on hazard-free code and $\geq$**0.6 IPC on CoreMark**.
- Verified commits against **Spike golden traces via RVFI**, passing all targeted instruction, forwarding, and hazard tests plus CoreMark.

Custom RISC-V Datapath and Standard-Cell VLSI Flow Cadence Virtuoso/Innovus, SystemVerilog

- Designed custom CMOS standard cells and a bit-sliced **RV32I datapath** through layout, Liberty/LEF views, and Innovus P&R.
- Verified cells and datapath via functional simulation, **DRC/LVS**, and extracted-view comparison against SystemVerilog golden outputs.

Experience

Design Verification Intern Jan. 2026 – Present
VortexNet LLC

- Verified a 4×4 **INT8 matrix-multiply datapath** with programmable scaling, ReLU, saturation, and valid/ready streaming control.
- Built a UVM testbench with constrained-random stimulus, a bit-accurate reference model, scoreboard, SVA, and random backpressure.
- Reached **100% functional coverage** across arithmetic and interface corner cases; debugged RTL failures with simulation logs and Verdi.

Undergraduate Research Assistant Jan. 2026 – May 2026
UIUC PlatformX Group

- Built **Python tools** to analyze custom NPU/TPU simulator outputs and characterize LLM inference across hardware configs.
- Analyzed **latency, throughput, HBM bandwidth, MXU utilization, and arithmetic intensity** to identify bottlenecks.